

Model Testing — Tab Field Reference

Covers the 6 tabs on the **Model Testing** page ([Page3Inference.jsx](#)): Transactions, Analytics, Credit Score, Anomalies, Model Evaluation, BRE payload.

All data comes from one API call, `POST /api/inference/run`, built by `_build_bundle()` in `BRE-backend/app/routers/inference.py`. If a bank statement was uploaded for the selected data source, every field below is computed for real from the parsed statement (`dataSource: "UPLOADED_STATEMENT"`). Otherwise everything is deterministically simulated from a seed derived from the **Custom ID** (`dataSource: "SIMULATED"`) — same Custom ID always reproduces the same numbers, but there is no real statement behind it.

1. Transactions tab

One row per transaction, from `bundle.transactions` (`generate_transactions()` / `map_real_transactions()` in `inference_engine.py`).

Field	Meaning
<code>date</code>	Transaction date (ISO <code>YYYY-MM-DD</code>). Simulated data spans a synthetic 12-entry timeline starting Feb 2026; real data uses the statement's own dates.
<code>narration</code>	Raw transaction description/memo (e.g. "NEFT STARK RETAIL SALARY", "ATM CASH WITHDRAWAL").
<code>type</code>	<code>CREDIT</code> (money in) or <code>DEBIT</code> (money out).
<code>amount</code>	Transaction amount in ₹, 2 decimal places.
<code>category</code>	Simulated data: one of Salary, Rent, Loan Repayment, Food, Utilities, Shopping, Entertainment, Tax, Cash Withdrawal, Travel, Revenue (from a fixed template list). Real data: <code>Bounce</code> (narration matches a bounce/return/dishonour/insufficient-funds pattern), <code>Cash Withdrawal</code> (matches ATM/cash-withdrawal pattern), or <code>Other</code> .
<code>merchant</code>	Always - (not populated by either path).

Field	Meaning
<code>stage</code>	<code>RULE</code> for simulated transactions (generated by template), <code>REAL</code> for transactions parsed from an uploaded statement.
<code>confidence</code>	0–100 confidence bar. Simulated: random 88–99. Real: always 100 (it's the literal parsed value, not a prediction).

2. Analytics tab

`bundle.analytics` (`generate_analytics()`), one 12-month chart + table **per selected model** — the shape of `chart/tableRows` differs by `modelId`.

Common wrapper fields

Field	Meaning
<code>modelId / name</code>	Which sub-model this analytics view belongs to.
<code>badge</code>	One-line status pill, e.g. "Low Risk (Approved)", "Healthy Cashflow (12.3K Net)", "Clean Account (Anomaly Index 4)", "Stable Balance (ADB ₹185,000)".
<code>metric</code>	Headline metric string shown next to the badge (e.g. "Credit Score: 712 / 900 PD: 3.2%").
<code>chartTitle</code>	Chart heading, includes the deployed model version.
<code>chartType</code>	<code>area</code> or <code>bar</code> (controls which Recharts component renders).
<code>dataKey / unit / chartColor / yDomain</code>	Recharts plumbing — which numeric field drives the chart, its unit suffix, line/bar color, and (risk model only) a fixed 0–10 y-axis.
<code>tableColumns</code>	Column headers for the table below the chart (model-specific, see below).
<code>chart</code>	Array of 12 monthly points, fields vary by model (below).
<code>tableRows</code>	Array of 12 rows, generically named <code>col1..col7</code> — meanings follow the <code>tableColumns</code> order for that model.

Risk Model

- Chart fields: **month**, **PDRiskPct** (probability of default, %), **CreditScore** (300–900).
- Table columns: Month, Default Risk (PD %), Credit Score (300-900), Inflow (₹), Outflow (₹), Avg Daily Balance (ADB), Risk Grade.
- The final month is anchored to the same score as the Credit Score tab (real data) so the two tabs never disagree; earlier months are interpolated back from a lower starting point.

Cashflow Model

- Chart fields: **month**, **NetCashflow**, **Inflow**, **Outflow** (all ₹).
- Table columns: Month, Monthly Inflow (₹), Monthly Outflow (₹), Net Cashflow (₹), Cash Runway, DSCR Coverage, Health.
- Real data: **Inflow** is walked toward the statement's actual **avg_monthly_inflow** over the 12 months. **Outflow** is simulated as 60–68% of inflow.

Fraud Model

- Chart fields: **month**, **AnomalyIndex** (0–20, trending down over the year), **Velocity** (transactions/day, 280–520).
- Table columns: Month, Tx Volume, UPI Velocity / Day, CERSAI Check, Anomaly Score, Flagged Tx, Verdict.
- Entirely simulated (no real-data anchoring) — **AnomalyIndex** isn't derived from the uploaded statement.

Money Balance Model

- Chart fields: **month**, **ADBScore** (average daily balance, ₹), **MinBal** (minimum balance, ₹).
 - Table columns: Month, ADB (₹), Min Balance (₹), Peak Balance (₹), Volatility Index, NACH Bounces, Stability Rating.
 - Real data: **ADBScore** is walked toward the statement's actual minimum balance ÷ 0.10 (i.e. treats the real minimum as ~10% of ADB) over the 12 months.
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3. Credit Score tab (**risk_score**)

Two objects: **bundle.riskScore** and **bundle.brePayload.feature_vector**.

riskScore

Field	Meaning
score	Underwriting score, 300–900 (higher = better). Real data: compute_real_credit_score() — starts at 520 and is adjusted by DSCR, income stability, NACH bounces, cash-withdrawal ratio, negative balance, and transaction volatility. Simulated: a seeded random 480–900.
grade	LOW (score ≥ 700), MEDIUM (≥ 550), HIGH (< 550) risk.
pd	Probability of Default %, derived as $((900 - \text{score}) / 900) * 20$, floored at 0.4%.
decision	APPROVED (LOW), CONDITIONAL APPROVAL (MEDIUM), REJECTED (HIGH).

feature_vector (also reused in the BRE payload — see below)

Field	Meaning
account_age_days	Simulated: random 90–900 days. Real data: null — not derivable from a single statement.
avg_monthly_inflow	Real: total credits in the statement (assumes ~1-month/statement window). Simulated: random ₹35,000–62,000.
nach_bounce_count_90d	Count of transactions whose narration matches a bounce/return/dishonour/ECS-return pattern. Simulated: 1–5 if grade is HIGH, else 0.

Field	Meaning
<code>dscr_ratio</code>	Debt Service Coverage Ratio. Real: <code>total_credit / total_debit</code> , capped at 9.99. Simulated: random 1.1–2.7.
<code>cash_withdrawal_ratio</code>	Share of total debits that are cash withdrawals (ATM/cash-withdrawal narration match). Real: computed ratio. Simulated: random 0.01–0.12.
<code>transaction_volatility</code>	Coefficient of variation of transaction amounts (std dev / mean). Real: computed. Simulated: random 0.8–2.4.
<code>minimum_balance</code>	Real: lowest running balance seen in the statement. Simulated: random ₹5,000–30,000.
<code>foir_ratio</code>	Fixed Obligation to Income Ratio, $(total_debit / total_credit) * 100$.
<code>income_stability</code>	0–1 score from consistency of credit amounts ($1 - stddev/mean$ of credits, clamped). Simulated: random 0.7–0.99.

The page derives two lists from this data, not returned as separate API fields:

- **Positive Drivers** (client-side, `Page3Inference.jsx`): zero NACH bounces, $DSCR \geq 1.5x$, income stability $\geq 85\%$, cash withdrawal ratio $\leq 5\%$ — up to 3 shown.
- **Risk Monitoring Alerts** = `brePayload.negative_factors` (see BRE payload section).

4. Anomalies tab

`bundle.anomalies` (`generate_anomalies()`) — a subset (~30% of transactions, min 2) flagged as anomalous, ranked by score descending.

Field	Meaning
<code>date / narration / amount</code>	Copied from the underlying transaction.

Field	Meaning
score	Anomaly risk probability, 40.0%–95.0% (seeded random — not currently derived from the real IsolationForest model's actual score).
level	HIGH if score $\geq$ 70%, else MEDIUM.
reasons	One of UNUSUAL_AMOUNT, STATISTICAL_OUTLIER, VELOCITY_SPIKE, NEW_COUNTERPARTY (randomly assigned).

Note: unlike the Credit Score tab, this tab is *not* wired to real statement data even when one is uploaded — it always runs off the customId-seeded simulation (`generate_anomalies()` takes only `custom_id` + `transactions`, no `real_feature_vector`).

5. Model Evaluation tab

`bundle.evaluation(generate_evaluation())` — jittered around the selected model's catalog `baseAccuracy` (Risk 94.8%, Cashflow 92.4%, Fraud 98.9%, Money Balance 91.6% before any live training; live-trained models use their actual CV accuracy as the base instead).

evalMetrics

Field	Meaning
r2Score	Coefficient of determination, $\approx$ <code>baseAccuracy</code> $\pm$ 1%.
mse	Mean Squared Error, $\approx$ $(1 - \text{baseAccuracy}) \times 0.3\text{--}0.5$.
mae	Mean Absolute Error, $\approx$ <code>mse</code> $\times$ $0.6\text{--}0.75$.
precision	Positive Predictive Value %, $\approx$ <code>baseAccuracy</code> + $0\text{--}1.5\%$.
recall	Sensitivity / True Positive Rate %, $\approx$ <code>baseAccuracy</code> - $0\text{--}2\%$.
f1Score	Harmonic mean of precision & recall.

cvFolds

5 rows (`Fold 1..Fold 5`), each the same metrics as above with an extra small $\pm 1\%$ jitter per fold, plus `status` (always `PASSED`).

Note: clicking "Re-evaluate Cross Validation" re-seeds with a timestamp, so numbers change slightly on every re-run — it's illustrating stability, not re-running a real held-out evaluation each time (that only happens once, at training time, in `ml_trainer.py`).

6. BRE payload tab

`bundle.brePayload(generate_bre_payload())` — the full JSON a downstream Business Rules Engine would consume to make an underwriting decision.

Field	Meaning
<code>statement_id</code>	The Custom ID entered on the page.
<code>bank</code>	The Bank Name entered, or "Not Specified".
<code>total_transactions</code>	Count of transactions in <code>bundle.transactions</code> .
<code>status</code>	Always "ANALYZED".
<code>data_source</code>	"UPLOADED_STATEMENT" if a real parsed statement was used, else "SIMULATED".
<code>model_metadata.selected_model</code>	Display name of the active model (e.g. "Risk Model").
<code>model_metadata.version</code>	Deployed version tag (e.g. "v3.4").
<code>model_metadata.credit_score</code>	Same as <code>riskScore.score</code> .
<code>model_metadata.risk_grade</code>	Same as <code>riskScore.grade</code> .
<code>model_metadata.probability_of_default</code>	Same as <code>riskScore.pd</code> , but expressed as a 0–1 fraction ($\div 100$) instead of a percentage.
<code>feature_vector</code>	Identical object to the one described in the Credit Score section above.

Field	Meaning
<code>negative_factors</code>	Human-readable strings auto-generated when a feature crosses a risk threshold: cash withdrawal ratio > 8%, transaction volatility > 1.8, minimum balance < ₹10,000, FOIR > 40%, income stability < 0.85, any NACH bounces in 90 days. This list is what renders as "Risk Monitoring Alerts" on the Credit Score tab.

Cross-tab consistency

Real-data mode is deliberately anchored so numbers agree across tabs: `real_feature_vector` and `real_risk_score` (computed once in `_build_bundle()`) are threaded into `generate_analytics()` and `generate_bre_payload()`, so the Credit Score, Analytics, and BRE Payload tabs always describe the same underlying applicant. The two exceptions are **Anomalies** (always simulated) and **Model Evaluation** (always a jittered accuracy estimate, not a live re-evaluation) — see the notes above.